CS-A1150 TIETOKANNAT

Project part 2

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Changes made after the submission of the first part

We removed PRIMARY KEY status from many attributes in classes, after we found out its meaning. Also it did not give us create object with same value. For example in SparePart OrderID was PK, but it did not give us to add SparePart with other Partnumber, but with same OrderID.

Invoice class's attribute Sum was separated into original sum and new sum. Reminder has NewSum removed. It was done to calculate new sum of invoice by original sum + sum of Reminder.extra of OrderID.

ReservetionOfApiration got attributer EmployeID and DeviceID to simplify finding them.

All this changes do not change our plan from part 1, just simplify it.

CREATE TABLE statements

For names, statuses, license number and types we used CHAR. DATETIME for dates and INT for others. Invoice is a parent class for foreign keys cause it will be created anyway even if repair is not done, but regular maintenance is. Some primary keys are removed as classes have objects with same value in one column. OrderID is foreign key cause it connects all classes and there is no point to create some objects without not existing OrderID .

```
CREATE TABLE Reminder(
```

```
OrderID INT ,
Extra INT,
NewSum INT,
FOREIGN KEY (OrderID) REFERENCES Invoice(OrderID)
);

CREATE TABLE Customer(
CurstomerID INT PRIMARY KEY,
LicenNum CHAR(50),
Name CHAR(50),
Phone INT,
Mail VARCHAR(319),
Address CHAR(200),
FOREIGN KEY (LicenNum) REFERENCES Car(LicenNum)
);

CREATE TABLE Employe(
ID INT PRIMARY KEY,
Name CHAR(50),
AvabiliDateAndTime DATETIME
);


CREATE TABLE Invoice(
OrderID INT PRIMARY KEY,
CustomerID INT,
EmployeID INT,
Hours INT,
Deadline DATETIME,
```

```
Status CHAR(30),
OriginalSum INT,
NewSum INT DEFAULT OriginalSum,
RemindersNo INT DEFAULT 0
);

CREATE TABLE Car(
LicenNum CHAR(50) PRIMARY KEY,
Distance INT
);

CREATE TABLE SparePart(
OrderID INT,
Partnumber INT,
Amount INT,
Sum INT,
FOREIGN KEY (OrderID) REFERENCES Invoice(OrderID)
);

CREATE TABLE Reservation_of_operation(
OrderID INT ,
EmployeID INT,
DeviceID INT,
Starts DATETIME,
Ends DATETIME,
FOREIGN KEY (OrderID) REFERENCES Invoice(OrderID),
FOREIGN KEY (EmployeID) REFERENCES Employe(ID),
FOREIGN KEY (DeviceID) REFERENCES Device (DeviceID)
);
```

```
CREATE TABLE Absences(
EmployeID INT,
Starts DATETIME,
Ends DATETIME,
FOREIGN KEY (EmployeID) REFERENCES Employe (ID)
);

CREATE TABLE Repair(
RepairName CHAR(50),
OrderID INT ,
LicenNum CHAR(50),
DurationTotal INT,
FOREIGN KEY (LicenNum) REFERENCES Car(LicenNum),
FOREIGN KEY (OrderID) REFERENCES Invoice(OrderID)
);

CREATE TABLE Device(
DeviceID INT PRIMARY KEY,
DeviceType CHAR(50),
AvabilityDateAndTime DATETIME);

CREATE TABLE Operation(
Duration INT ,
OpreationName CHAR(50),
Distance INT,
ReqDevice CHAR(50) );

CREATE TABLE Regular_maintenance(
```

```
Distance INT ,
OrderID INT,
LicenNum CHAR(50) ,
DurationTotal INT,
FOREIGN KEY (LicenNum) REFERENCES Car(LicenNum),
FOREIGN KEY (OrderID) REFERENCES Invoice(OrderID));

CREATE TABLE Owner(
LicenNum CHAR(50),
Name CHAR(50),
FOREIGN KEY (LicenNum) REFERENCES Car(LicenNum)
);
```

/*Populate the table with values */

```
/*Add a new employe */
INSERT INTO Employe
VALUES(88, "James Creew", "01/01/2022 00:00:00:000");
/*[20:18:24] Query finished in 0.023 second(s). Rows affected: 1 */

/*Create new car */
INSERT INTO Car
Values( 'ABC-123', 40000);

INSERT INTO Car
Values( 'BVC-123', 80000);



/*Create new customer */
```

```
INSERT INTO Customer
Values(123, 'ABC-123', 'AlliSmith', 04081688, 'homer@home.com', 'Mistdrive 15');

INSERT INTO Customer
Values(456, 'BVC-123', 'Trudoro', 04088779, 'dseer@hpre.com', 'Doomdrive 98');


/* Create new Employe */
INSERT INTO Employe
VALUES(1000, 'Veeti Hakkinen', '2021-05-23T14:25:10' );
/*[21:17:25] Query finished in 0.013 second(s). Rows affected: 1 */

/* Create new Invoice */
INSERT INTO Invoice
VALUES(123, 123, 88, 48, "01/01/2022 00:00:00:000", "unpaid", 1000, 1000, 0);

UPDATE Invoice
SET RemindersNo = 1
WHERE OrderID = 123

INSERT INTO Invoice
VALUES(456, 456, 1000, 48, "04/01/2022 00:00:00:000", "unpaid", 1000, 1000, 0)

/* Create new reminder*/
INSERT INTO Reminder
VALUES(123, 320, 1320);

INSERT INTO Reminder
VALUES(456, 70, 1070);
```

```
INSERT INTO Reminder
VALUES(456, 70, 1140);


/*Creare new spare part*/
INSERT INTO SparePart
VALUES(123, 11427, 2, 70 );
/* [17:30:40] Query finished in 0.008 second(s). Rows affected: 1*/

/*Create new reservation for employee*/
INSERT INTO Reservation_of_operation
VALUES(123, 1000, NULL , "02/01/2022 00:00:00:000", "03/01/2022 00:07:00:000" );
/* [17:34:59] Query finished in 0.009 second(s). Rows affected: 1*/

/*Create new device */
INSERT INTO Device
VALUES(1, 'Hoister', "01/02/2022 00:00:00:000" );
/* [17:38:43] Query finished in 0.010 second(s). Rows affected: 1*/

/*Create new reservation for a device */
INSERT INTO Reservation_of_operation
VALUES(123, NULL, 1 , "02/01/2022 00:00:00:000", "03/01/2022 00:07:00:000" )
/* [17:41:15] Query finished in 0.007 second(s). Rows affected: 1*/

/*Create new absence of employee */
INSERT INTO Absences
VALUES(1000, "09/01/2022 00:00:00:000", "10/01/2022 00:07:00:000" );
```

```
/*Create new repair */
INSERT INTO Repair
VALUES("Engine swap", 123, "ABC-123", 7 );

/*Create new operations */
INSERT INTO Operation
VALUES(3, "Take engine out", NULL, "Hoister" );

INSERT INTO Operation
VALUES(4, "Put engine in", NULL, "Hoister" );

INSERT INTO Operation
VALUES(4, "Oil changet", 60000, NULL );


/*Create new owner*/
INSERT INTO Owner
VALUES("ABC-123", "Mary Holinen" );

INSERT INTO Owner
VALUES("ABC-123", "Jake Harin" );

/*Create new regular maintinance */
INSERT INTO Regular_maintenance
VALUES(70000, 456, "BVC-123",4);
```

/* Indices and views*/

CREATE INDEX DeviceType on Device(DeviceType);

*/*There are Devices with same type, but different ID. When we search for device for operation, we do not know ID, but we know type. Devices usually lives long enough, so table do not get updated very often.*/*

CREATE INDEX Reminders on Reminder(OrderID);

*/*There can be many unpaid Invoices and Reminder contains reminder info of each one. It speeds up searching for reminders of Invoices to calculate new sum. Usually deadline in long and Reminders will not get updated so often.*

Other classes get updated too often or are too small to be needed in indexes/*

CREATE VIEW HistoryOFLicenNum123 AS

SELECT OrderID

FROM Repair, Regular_maintenance

WHERE Car.LicenNum="ABC-123";

*/*Used to see all repairs of car */*

CREATE VIEW PartsForOrder123 AS

SELECT *

FROM SparePart

WHERE OrderID=123;

Used to see all parts used in work on specific order.

/* SQL queries for typical use cases */

/* Find all employes available between given dates to place him at work*/

SELECT *

FROM Employe

WHERE AvabiliDateAndTime BETWEEN '01/01/2022 00:00:00:000' AND '01/03/2022 00:00:00.000';

	ID	Name	AvabiliDateAndTime
1	88	James Creew	01/01/2022 00:00:00:000

Status

[17:50:40] Error while executing SQL query on data

[17:50:44] Query finished in 0.007 second(s). Rows

[17:51:09] Query finished in 0.002 second(s).

/*Find contacts of customer who brought the car to contact him about repair*/

SELECT Name, Phone, Mail

FROM Customer

WHERE LicenNum = "ABC-123"

	Name	Phone	Mail
1	AlliSmith	4081688	homer@home.com

Status

[17:51:09] Query finished in 0.002 second(s).

[17:52:33] Error while executing SQL query on d

[17:52:48] Query finished in 0.001 second(s).

/*Find all operations needed for current distance for regular maintenance*/

SELECT *

FROM Operation, (Select * FROM Regular_maintenance WHERE OrderID=456) AS X

WHERE Operation.Distance <= X.Distance;

	Duration	OpreationN	Distance	ReqDevice	Distance:1	OrderID	LicenNum	DurationT
1	4	Oil changet	60000	NULL	70000	456	BVC-123	4

Status

[18:05:21] Error while executing SQL query on database 'dsa': near "WHERE": syntax error

[18:04:33] Query finished in 0.004 second(s).

[18:04:45] Query finished in 0.004 second(s).

/*Find all parts used in work to add them to bill*/

SELECT *

FROM SparePart

WHERE OrderID=123;

	OrderID	Partnumber	Amount	Sum
1	123	11427	2	70

Status

[18:04:45] Query finished in 0.004 second(s).

[18:05:20] Error while executing SQL query on

[18:05:28] Query finished in 0.002 second(s).

/*Find all Reminders of Invoice to calculate new sum*/

SELECT *

FROM Reminder

WHERE OrderID=123;

	OrderID	Extra	NewSum
1	123	320	1320

Status

[18:05:20] Error while executing SQL query on

[18:05:28] Query finished in 0.002 second(s).

[18:06:05] Query finished in 0.001 second(s).

/*Find customers with unpaid invoices to send reminders*/

SELECT Name, Phone, Mail, Address

FROM Customer JOIN Invoice ON Invoice.CustomerID=Customer.CurstomerID

WHERE Invoice.status = "unpaid";

	Name	Phone	Mail	Address
1	AlliSmith	4081688	homer@home.com	Mistdrive 15
2	Trudoro	4088779	dseer@hpre.com	Doomdrive 98

Status

[18:06:38] Error while executing SQL query on database 'd

[18:07:20] Query finished in 0.003 second(s).

/*Count number of reminders for each Invoice to make statistic*/

SELECT OrderID, COUNT(OrderID)

FROM Reminder

GROUP BY OrderID

HAVING COUNT (OrderID) >=1;

	OrderID	COUNT(OrderID)
1	123	1
2	456	2

Status

[18:15:24] Query finished in 0.011 second(s). Ro

[18:15:44] Query finished in 0.002 second(s).

/* Find all owners of car if police asks*/

SELECT Name

FROM Car JOIN Owner ON Car.LicenNum=Owner.LicenNum

WHERE Car.LicenNum="ABC-123";

	Name
1	Mary Holinen
2	Jake Harin

Status

[18:17:28] Query finished in 0.001 second(s).

[18:18:12] Query finished in 0.007 second(s). Rc

[18:18:48] Query finished in 0.001 second(s).

/*Find all unpaid invoices to send reminders*/

SELECT *

FROM Invoice

WHERE status="unpaid";

	OrderID	Customer	EmployeI[	Hours	Deadline	Status	OriginalSu	NewSum	Reminder
1	123	123	88	48	01/01/2022 00:00:00:000	unpaid	1000	1000	1
2	456	456	1000	48	04/01/2022 00:00:00:000	unpaid	1000	1000	0

Status

[18:18:12] Query finished in 0.007 second(s). Rows affected: 1

[18:18:48] Query finished in 0.001 second(s).

[18:19:19] Query finished in 0.003 second(s).

/*Find device with type available between dates to book it for worker*/

SELECT *

FROM Device

WHERE DeviceType="Hoister" AND AvabilityDateAndTime BETWEEN "01/01/2022 00:00:00:000" AND "01/03/2022 00:00:00.000";

	DeviceID	DeviceType	AvabilityDateAndTime
1	1	Hoister	01/02/2022 00:00:00:000

Status

[18:19:19] Query finished in 0.003 second(s).

[18:20:08] Query finished in 0.001 second(s).

[18:20:57] Query finished in 0.001 second(s).

/*Find all Invoices with deadline today to send reminders*/

SELECT *

FROM Invoice

WHERE Deadline= "01/01/2022 00:00:00:000";

	OrderID	Customer	EmployeII	Hours	Deadline	Status	OriginalSu	NewSum	Reminder
1	123	123	88	48	01/01/2022 00:00:00:000	unpaid	1000	1000	0

Status

[18:18:14] Query finished in 0.001 second(s).

[18:18:37] Query finished in 0.001 second(s).

[18:19:32] Error while executing SQL query on database 'Garage': no such function: GetDate

[18:20:33] Query finished in 0.002 second(s).

/* Find all Absences of employe to see how good worker he is*/

Select*

From Absences

Where EmployeID=1000;

Grid view | Form view

Total rows loaded: 1

	EmployeID	Starts	Ends
1	1000	09/01/2022 00:00:00:000	10/01/2022 00:07:00:000

Status

[18:16:15] Query finished in 0.002 second(s).

[18:17:37] Error while executing SQL query on database 'Garage': no such column: EmpoyeID

[18:18:14] Query finished in 0.001 second(s).

[18:18:37] Query finished in 0.001 second(s).

/*Find which employees are working on a given order so they get paid*/

SELECT *

FROM Employe JOIN Reservation_of_operation ON Employe.ID = Reservation_of_operation.EmployeID

WHERE Reservation_of_operation.OrderID = 123

	ID	Name	AvabiliDateAndTime	OrderID	EmployeI	DeviceID	Starts	Ends
1	1000	Veeti Hakkinen	2021-05-23T14:25:10	123	1000	NULL	02/01/2022 00:00:00:000	03/01/2022 00:07:00:000

Status

[18:10:57] Query finished in 0.002 second(s).

[18:13:02] Query finished in 0.003 second(s).

[18:14:56] Error while executing SQL query on database 'Garage': no such column: Reservation_of_operation.EmployeID

[18:16:15] Query finished in 0.002 second(s).

/* Find which devices are reserved on in a given time interval to find them if they aren't in storage*/

SELECT *

FROM Device JOIN Reservation_of_operation ON Device.DeviceID = Reservation_of_operation.DeviceID

WHERE Reservation_of_operation.Starts ="02/01/2022 00:00:00:000"

AND Reservation_of_operation.Ends= "03/01/2022 00:07:00:000"

	DeviceID	DeviceTyp	AvabilityDateAndTime	OrderID	EmployeII	DeviceID:	Starts	Ends
1	1	Hoister	01/02/2022 00:00:00:000	123	NULL	1	02/01/2022 00:00:00:000	03/01/2022 00:07:00:000

Status

[18:10:08] Query finished in 0.002 second(s).

[18:10:21] Query finished in 0.002 second(s).

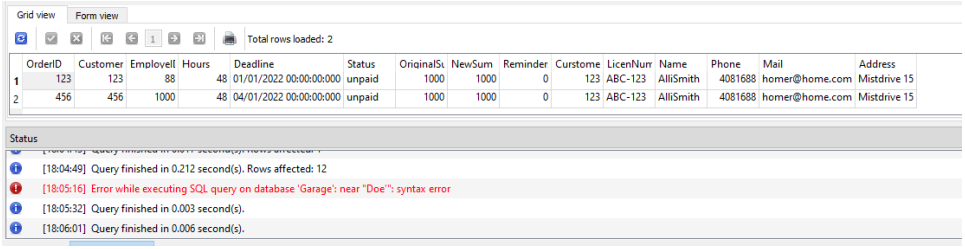
[18:10:57] Query finished in 0.002 second(s).

[18:13:02] Query finished in 0.003 second(s).

Movies (data) | SQL editor 2

/* Find all invoices for a given owner (all invoices for all cars own by a given person) to see how good client he is*/

```
SELECT *

FROM Invoice JOIN Customer ON CustomerID

WHERE Customer.LicenNum =

(SELECT LicenNum

FROM Owner

WHERE Owner.Name = "Mary Holinen");
```



Grid view | Form view

Total rows loaded: 2

	OrderID	Customer	EmployeII	Hours	Deadline	Status	OriginalSu	NewSum	Reminder	Curstome	LicenNum	Name	Phone	Mail	Address
1	123	123	88	48	01/01/2022 00:00:00:000	unpaid	1000	1000	0	123	ABC-123	AlliSmith	4081688	homer@home.com	Mistdrive 15
2	456	456	1000	48	04/01/2022 00:00:00:000	unpaid	1000	1000	0	123	ABC-123	AlliSmith	4081688	homer@home.com	Mistdrive 15

Status

[18:04:49] Query finished in 0.212 second(s). Rows affected: 12

[18:05:16] Error while executing SQL query on database 'Garage': near "Doe'": syntax error

[18:05:32] Query finished in 0.003 second(s).

[18:06:01] Query finished in 0.006 second(s).